

Verification Plan Report

Project: AXI-Stream Verification

Version: 1.6

Verification Role: Master

1. Test Requirements

Index	Feature/Sub	Name	Description
TR_HS_01	Handshake (Stability)	Master Stability	Master shall not toggle TDATA/TSTRB/TLAST/TID/TDEST until TREADY is HIGH.
TR_HS_02	Handshake (Independent)	No TREADY Waiting	Master must not wait for TREADY before asserting TVALID.
TR_RST_01	Reset (Suppression)	Reset TVALID LOW	Master must drive TVALID LOW during reset.
TR_RST_02	Reset (Synchronous)	Sync Recovery	Master can only drive TVALID 1 cycle after ARESETn deassertion.
TR_PKT_01	Packet (Identity)	ID Consistency	TID must remain constant within a packet.
TR_PKT_02	Packet (Identity)	DEST Consistency	TDEST must remain constant within a packet.
TR_QUAL_01	Qualifiers (TKEEP)	Null Bytes	TKEEP identifies valid data bytes in a beat.
TR_QUAL_02	Qualifiers (TSTRB)	Strobes	TSTRB identifies data vs position bytes.
TR_PWR_01	Power (AXI5)	TWAKEUP Rule	TWAKEUP must indicate activity associated with the interface.
TR_PWR_02	Power (AXI5)	Wakeup Stability	If TVALID and TWAKEUP are HIGH, TWAKEUP must stay HIGH until TREADY.

2. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_HS_01	Valid before Ready: Master drives data, Slave stalls 10 cycles. Verify stability.	TR_HS_01, TR_HS_02
TC_HS_02	Ready before Valid: Slave waiting. Master asserts valid. verify 1 cycle handshake.	TR_HS_01
TC_HS_03	Back-to-back: Master drives infinite stream with TREADY=1.	TR_HS_02
TC_RST_01	Async Reset: Pulse ARESETn during transfer. Verify TVALID drops.	TR_RST_01
TC_RST_02	Fast Recovery: Attempt TVALID on cycle 0 of reset release (Invalid).	TR_RST_02

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TC_PKT_01	Standard Packet: 16-beat transfer with TLAST at end.	TR_PKT_01, TR_PKT_02
TC_PKT_02	Zero-Byte Packet: TLAST=1 with all TKEEP=0.	TR_PKT_01
TC_PKT_03	ID Change mid-packet: Change TID while TLAST is LOW. Verify assertion.	TR_PKT_01
TC_PKT_04	DEST Switch: Change TDEST while TLAST is LOW. Verify assertion.	TR_PKT_02
TC_QUAL_01	Sparse TSTRB: Transfer with only byte 0 and byte 3 strobed.	TR_QUAL_02
TC_QUAL_02	Partial TKEEP: Transfer with null bytes interspersed.	TR_QUAL_01
TC_PWR_01	AXI5 Wakeup: TWAKEUP asserted 2 cycles before TVALID.	TR_PWR_01
TC_PWR_02	Wakeup Violation: TWAKEUP drops while TVALID is waiting for TREADY.	TR_PWR_02
TC_SIDE_01	TUSER stability: Change TUSER mid-transfer before TREADY.	TR_HS_01
TC_SIDE_02	TUSER mapping: Verify sideband bits match scoreboard.	TR_HS_01

3. Functional Coverage Groups

Group Name	Description
cg_handshake	Cross of TVALID/TREADY including stalls.
cg_packet	Coverage of TLAST transitions and ID switches.
cg_qualifiers	Exhaustive strobe and keep patterns.

4. Interface Signals

Signal Name	Direction	Width	Description
TDATA	Master	TDATA_WIDTH	Payload
TVALID	Master	1	Valid signal
TREADY	Slave	1	Ready signal
TLAST	Master	1	Packet End
TID	Master	TID_WIDTH	Stream ID
TDEST	Master	TDEST_WIDTH	Dest ID
TKEEP	Master	TD_W/8	Null Byte Marker
TSTRB	Master	TD_W/8	Byte Strobe
TWAKEUP	Master	1	Activity Info
ARESETn	Global	1	Active LOW Reset